MIL-R-22684/2B
21 November 1968
SUPERSEDING
MIL-R-22684/2A
20 January 1965

MILITARY SPECIFICATION SHEET

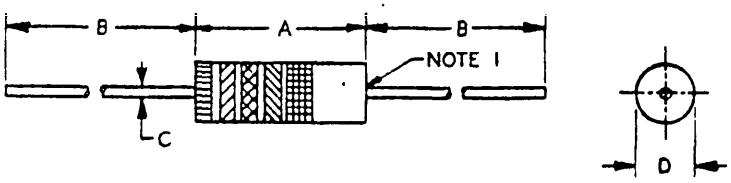
RESISTORS, FIXED, FILM, INSULATED,

STYLE RL20

(B) INACTIVE FOR DESIGN AFTER DATE OF THIS SPECIFICATION SHEET (SEE PAGE 8). USE MIL-R-39017/2.

The complete requirements for procuring the resistors described herein shall consist of this document and the latest issue of Specification MIL-R-22684.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.335	.416	8.74	10.57
B	1.375	1.625	34.92	41.28
C	.026	.036	.66	.91
D	.115	.161	2.92	4.09

NOTES:

1. The end of the body shall be that point at which the diameter equals the nearest drill size larger than 250 percent of the nominal lead diameter. The leads shall be solderable to within .125 inch (3.18 mm) of the resistor body.

2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

(B) denotes changes

FSC 5905

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REQUIREMENTS:

Dimensions and configuration: See figure.

Resistance values: See table I.

Resistance tolerances: See table I.

Terminal type: See table I.

Minimum resistance value: 4.3 ohms.

Maximum resistance value: 0.47 megohm.

Maximum continuous working voltage: 350 volts.

Power rating: 1/2 watt at 70°C.

Temperature cycling: Method 102, test condition D, MIL-STD-202. Change in resistance shall not exceed ±(1 percent + 0.05 ohm).

Low temperature operation:
Change in resistance shall not exceed ±(0.5 percent + 0.05 ohm).

Short time overload:
Change in resistance shall not exceed ±(0.5 percent + 0.05 ohm).

Terminal strength: Method 211 of MIL-STD-202.
Test condition A - Pull test - 5 pounds.
Test condition D - Twist test - Rotated 360° for 3 rotations. Change in resistance shall not exceed ±(0.5 percent + 0.05 ohm).

Dielectric withstanding voltage:
Methods 301 and 105 of MIL-STD-202.
Atmospheric pressure - 700 volts rms.
Barometric pressure - 400 volts rms.

Insulation resistance:
Method 302 of MIL-STD-202. Test condition A or B, whichever is practicable. Insulation resistance shall be not less than 100 megohms.

Resistance to soldering heat:
Method 210, test condition A, MIL-STD-202. Change in resistance shall not exceed ±(0.5 percent +0.05 ohm).

Moisture resistance:
Method 106 of MIL-STD-202. Change in resistance shall not exceed 1.5 percent. Insulation resistance shall be 100 megohms minimum.

Resistance temperature characteristic:
Method 304 of MIL-STD-202. Change in resistance shall not exceed ±0.020 percent per degree C.

Life: Method 108, test condition D, MIL-STD-202. Change in resistance shall not exceed 2.0 percent.

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TABLE I. Part number designation.

Type designation	Nominal total resistance value	Resistance tolerance (Percent)	Terminal	Dash number
	Ohms			
RI20S4R3G	4.3	G	S	0227
RI20S4R3J	4.3	J	S	0228
RI20S4R7G	4.7	G	S	0229
RI20S4R7J	4.7	J	S	0230
RI20S5R1G	5.1	G	S	0231
RI20S5R1J	5.1	J	S	0232
RI20S5R6G	5.6	G	S	0233
RI20S5R6J	5.6	J	S	0234
RI20S6R2G	6.2	G	S	0235
RI20S6R2J	6.2	J	S	0236
RI20S6R8G	6.8	G	S	0237
RI20S6R8J	6.8	J	S	0238
RI20S7R5G	7.5	G	S	0239
RI20S7R5J	7.5	J	S	0240
RI20S8R2G	8.2	G	S	0241
RI20S8R2J	8.2	J	S	0242
RI20S9R1G	9.1	G	S	0243
RI20S9R1J	9.1	J	S	0244
RI20S100G	10	G	S	0001
RI20S100J	10	J	S	0002
RI20S110G	11	G	S	0003
RI20S110J	11	J	S	0004
RI20S120G	12	G	S	0005
RI20S120J	12	J	S	0006
RI20S130G	13	G	S	0007
RI20S130J	13	J	S	0008
RI20S150G	15	G	S	0009
RI20S150J	15	J	S	0010
RI20S160G	16	G	S	0011
RI20S160J	16	J	S	0012
RI20S180G	18	G	S	0013
RI20S180J	18	J	S	0014
RI20S200G	20	G	S	0015
RI20S200J	20	J	S	0016
RI20S220G	22	G	S	0017
RI20S220J	22	J	S	0018
RI20S240G	24	G	S	0019
RI20S240J	24	J	S	0020
RI20S270G	27	G	S	0021
RI20S270J	27	J	S	0022
RI20S300G	30	G	S	0023
RI20S300J	30	J	S	0024
RI20S330G	33	G	S	0025
RI20S330J	33	J	S	0026
RI20S360G	36	G	S	0027
RI20S360J	36	J	S	0028

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TABLE I. Part number designation. (Cont)

Type designation	Nominal total resistance value	Resistance tolerance (Percent)	Terminal	Dash number
	Ohms			
RI20S390G	39	G	S	0029
RI20S390J	39	J	S	0030
RI20S430G	43	G	S	0031
RI20S430J	43	J	S	0032
RI20S470G	47	G	S	0033
RI20S470J	47	J	S	0034
RI20S510G	51	G	S	0035
RI20S510J	51	J	S	0036
RI20S560G	56	G	S	0037
RI20S560J	56	J	S	0038
RI20S620G	62	G	S	0039
RI20S620J	62	J	S	0040
RI20S680G	68	G	S	0041
RI20S680J	68	J	S	0042
RI20S750G	75	G	S	0043
RI20S750J	75	J	S	0044
RI20S820G	82	G	S	0045
RI20S820J	82	J	S	0046
RI20S910G	91	G	S	0047
RI20S910J	91	J	S	0048
RI20S101G	100	G	S	0049
RI20S101J	100	J	S	0050
RI20S111G	110	G	S	0051
RI20S111J	110	J	S	0052
RI20S121G	120	G	S	0053
RI20S121J	120	J	S	0054
RI20S131G	130	G	S	0055
RI20S131J	130	J	S	0056
RI20S151G	150	G	S	0057
RI20S151J	150	J	S	0058
RI20S161G	160	G	S	0059
RI20S161J	160	J	S	0060
RI20S181G	180	G	S	0061
RI20S181J	180	J	S	0062
RI20S201G	200	G	S	0063
RI20S201J	200	J	S	0064
RI20S221G	220	G	S	0065
RI20S221J	220	J	S	0066
RI20S241G	240	G	S	0067
RI20S241J	240	J	S	0068
RI20S271G	270	G	S	0069
RI20S271J	270	J	S	0070
RI20S301G	300	G	S	0071
RI20S301J	300	J	S	0072
RI20S331G	330	G	S	0073
RI20S331J	330	J	S	0074
RI20S361G	360	G	S	0075
RI20S361J	360	J	S	0076

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TABLE I. Part number designation. (Cont)

Type designation	Nominal total resistance value	Resistance tolerance (Percent)	Terminal	Dash number
	Ohms			
RI20S391G	390	G	S	0077
RI20S391J	390	J	S	0078
RI20S431G	430	G	S	0079
RI20S431J	430	J	S	0080
RI20S471G	470	G	S	0081
RI20S471J	470	J	S	0082
RI20S511G	510	G	S	0083
RI20S511J	510	J	S	0084
RI20S561G	560	G	S	0085
RI20S561J	560	J	S	0086
RI20S621G	620	G	S	0087
RI20S621J	620	J	S	0088
RI20S681G	680	G	S	0089
RI20S681J	680	J	S	0090
RI20S751G	750	G	S	0091
RI20S751J	750	J	S	0092
RI20S821G	820	G	S	0093
RI20S821J	820	J	S	0094
RI20S911G	910	G	S	0095
RI20S911J	910	J	S	0096
RI20S102G	1,000	G	S	0097
RI20S102J	1,000	J	S	0098
RI20S112G	1,100	G	S	0099
RI20S112J	1,100	J	S	0100
RI20S122G	1,200	G	S	0101
RI20S122J	1,200	J	S	0102
RI20S132G	1,300	G	S	0103
RI20S132J	1,300	J	S	0104
RI20S152G	1,500	G	S	0105
RI20S152J	1,500	J	S	0106
RI20S162G	1,600	G	S	0107
RI20S162J	1,600	J	S	0108
RI20S182G	1,800	G	S	0109
RI20S182J	1,800	J	S	0110
RI20S202G	2,000	G	S	0111
RI20S202J	2,000	J	S	0112
RI20S222G	2,200	G	S	0113
RI20S222J	2,200	J	S	0114
RI20S242G	2,400	G	S	0115
RI20S242J	2,400	J	S	0116
RI20S272G	2,700	G	S	0117
RI20S272J	2,700	J	S	0118
RI20S302G	3,000	G	S	0119
RI20S302J	3,000	J	S	0120
RI20S332G	3,300	G	S	0121
RI20S332J	3,300	J	S	0122
RI20S362G	3,600	G	S	0123
RI20S362J	3,600	J	S	0124
RI20S392G	3,900	G	S	0125
RI20S392J	3,900	J	S	0126

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TABLE I. Part number designation. (Cont)

Type designation	Nominal total resistance value	Resistance tolerance (Percent)	Terminal	Dash number
	Ohms			
RL20S432G	4,300	G	S	0127
RL20S432J	4,300	J	S	0128
RL20S472G	4,700	G	S	0129
RL20S472J	4,700	J	S	0130
RL20S512G	5,100	G	S	0131
RL20S512J	5,100	J	S	0132
RL20S562G	5,600	G	S	0133
RL20S562J	5,600	J	S	0134
RL20S622G	6,200	G	S	0135
RL20S622J	6,200	J	S	0136
RL20S682G	6,800	G	S	0137
RL20S682J	6,800	J	S	0138
RL20S752G	7,500	G	S	0139
RL20S752J	7,500	J	S	0140
RL20S822G	8,200	G	S	0141
RL20S822J	8,200	J	S	0142
RL20S912G	9,100	G	S	0143
RL20S912J	9,100	J	S	0144
RL20S103G	10,000	G	S	0145
RL20S103J	10,000	J	S	0146
RL20S113G	11,000	G	S	0147
RL20S113J	11,000	J	S	0148
RL20S123G	12,000	G	S	0149
RL20S123J	12,000	J	S	0150
RL20S133G	13,000	G	S	0151
RL20S133J	13,000	J	S	0152
RL20S153G	15,000	G	S	0153
RL20S153J	15,000	J	S	0154
RL20S163G	16,000	G	S	0155
RL20S163J	16,000	J	S	0156
RL20S183G	18,000	G	S	0157
RL20S183J	18,000	J	S	0158
RL20S203G	20,000	G	S	0159
RL20S203J	20,000	J	S	0160
RL20S223G	22,000	G	S	0161
RL20S223J	22,000	J	S	0162
RL20S243G	24,000	G	S	0163
RL20S243J	24,000	J	S	0164
RL20S273G	27,000	G	S	0165
RL20S273J	27,000	J	S	0166
RL20S303G	30,000	G	S	0167
RL20S303J	30,000	J	S	0168
RL20S333G	33,000	G	S	0169
RL20S333J	33,000	J	S	0170
RL20S363G	36,000	G	S	0171
RL20S363J	36,000	J	S	0172
RL20S393G	39,000	G	S	0173
RL20S393J	39,000	J	S	0174
RL20S433G	43,000	G	S	0175
RL20S433J	43,000	J	S	0176

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TABLE I. Part number designation. (Cont)

Type designation	Nominal total resistance value	Resistance tolerance (Percent)	Terminal	Dash number
	Ohms			
RI20S473G	47,000	G	S	0177
RI20S473J	47,000	J	S	0178
RI20S513G	51,000	G	S	0179
RI20S513J	51,000	J	S	0180
RI20S563G	56,000	G	S	0181
RI20S563J	56,000	J	S	0182
RI20S623G	62,000	G	S	0183
RI20S623J	62,000	J	S	0184
RI20S683G	68,000	G	S	0185
RI20S683J	68,000	J	S	0186
RI20S753G	75,000	G	S	0187
RI20S753J	75,000	J	S	0188
RI20S823G	82,000	G	S	0189
RI20S823J	82,000	J	S	0190
RI20S913G	91,000	G	S	0191
RI20S913J	91,000	J	S	0192
	Megohm			
RI20S104G	0.10	G	S	0193
RI20S104J	0.10	J	S	0194
RI20S114G	0.11	G	S	0195
RI20S114J	0.11	J	S	0196
RI20S124G	0.12	G	S	0197
RI20S124J	0.12	J	S	0198
RI20S134G	0.13	G	S	0199
RI20S134J	0.13	J	S	0200
RI20S154G	0.15	G	S	0201
RI20S154J	0.15	J	S	0202
RI20S164G	0.16	G	S	0203
RI20S164J	0.16	J	S	0204
RI20S184G	0.18	G	S	0205
RI20S184J	0.18	J	S	0206
RI20S204G	0.20	G	S	0207
RI20S204J	0.20	J	S	0208
RI20S224G	0.22	G	S	0209
RI20S224J	0.22	J	S	0210
RI20S244G	0.24	G	S	0211
RI20S244J	0.24	J	S	0212
RI20S274G	0.27	G	S	0213
RI20S274J	0.27	J	S	0214
RI20S304G	0.30	G	S	0215
RI20S304J	0.30	J	S	0216
RI20S334G	0.33	G	S	0217
RI20S334J	0.33	J	S	0218
RI20S364G	0.36	G	S	0219
RI20S364J	0.36	J	S	0220
RI20S394G	0.39	G	S	0221
RI20S394J	0.39	J	S	0222
RI20S434G	0.43	G	S	0223
RI20S434J	0.43	J	S	0224
RI20S474G	0.47	G	S	0225
RI20S474J	0.47	J	S	0226

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Shock, medium impact: Method 205, test condition C, MIL-STD-202. Change in resistance shall not exceed ±(0.5 percent +0.05 ohm).

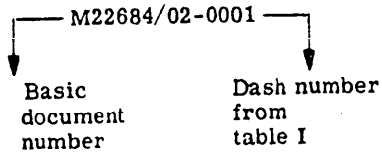
Vibration, high frequency: Method 204, test condition D, MIL-STD-202. Change in resistance shall not exceed ±(0.5 percent + 0.05 ohm).

Solderability: Method 208 of MIL-STD-202.

Ⓑ Substitution data. The resistors specified herein are not for use in design after date of this specification sheet. They are suitable for use in design contracts current to that date and to support existing Military equipment. Resistors specified in MIL-R-39017/2 are preferred for design and regardless of the failure rate designation can be used as substitutes for the inactivated resistors of the same resistance value and tolerance, of this specification.

Part number: Consists of the basic number of this specification sheet and a dash number taken from table I.

Example:



Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU, MI
Navy - EC
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - AV
Navy -
Air Force - 19

Preparing activity:
Army - EL

(Project 5905-0657-4)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-R-22684/2B Resistors, Fixed, Film, Insulated Style RL20

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A. IF SO, PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity - Optional) | DATE

DD FORM 1426 REPLACES EDITION OF OCT 54 WHICH MAY BE USED.
D30-FM 1068-68

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U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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